Chemical Identity Lost in Regulation: A Study of Semantic Interoperability in European Chemical Substance Data

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Abstract. Chemical substances are fundamental entities in environmental regulation, yet their representation in regulatory data systems diverges significantly from established practices in chemistry and cheminformatics. While chemical identity can be unambiguously defined using structure-based identifiers such as the International Chemical Identifier (InChI) and InChIKey, European regulatory datasets rely on heterogeneous identifiers including CAS numbers, EC numbers, textual names, and loosely defined group descriptions.

This paper analyses the resulting interoperability challenges across European regulatory frameworks, based on a case study spanning 18 datasets from the European Chemicals Agency (ECHA) and related sources. We show that regulatory "substances" frequently represent heterogeneous semantic entities, including individual molecules, mixtures, analytical parameters, and substance groups, for which formal relationships are not explicitly defined.

Our results demonstrate that (i) a substantial fraction of regulatory entries cannot be linked to a defined chemical structure, (ii) CAS numbers do not provide a reliable unique identifier, and (iii) semantic and embedding-based approaches fail to recover chemical identity for non-structure substances. We further show that resolving these inconsistencies retrospectively leads to combinatorial complexity that is infeasible at scale.

In contrast, structure-based identifiers and ontology-driven approaches, as used in cheminformatics resources such as PubChem and ChEBI, enable scalable integration through linked data principles. We argue that interoperability of chemical substance data must be addressed at the level of legislation and data modelling, by adopting structure-based identifiers, formal ontologies, and machine-readable representations.

Keywords. chemical substance identification, semantic interoperability, regulatory interoperability, REACH, InChIKey, cheminformatics, FAIR data, knowledge graph, European environmental legislation

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1. Introduction

Chemical substances are central entities in environmental, health, and product regulation. Across European legislation, substances are used to define obligations, assess risks, and communicate hazards in frameworks such as REACH [1], CLP [2], and sector-specific regulations including pesticides and water quality directives. Effective governance of these substances increasingly depends on the ability to integrate and reuse data across regulatory domains.

In chemistry and cheminformatics, chemical identity is well-defined and operationalised through structure-based representations. Molecular structures can be uniquely encoded using standards such as the International Chemical Identifier (InChI) and its hashed representation, the InChIKey, enabling unambiguous identification, comparison, and linking of substances across databases. Large-scale infrastructures such as PubChem and ChEBI build on these identifiers to provide machine-readable, ontology-based knowledge graphs that support interoperability and automated reasoning.

In contrast, regulatory data systems adopt a fundamentally different approach. Substances are typically identified using heterogeneous and often ambiguous identifiers, including CAS registry numbers, EC numbers, textual names, and informal group descriptions. Moreover, the concept of a "substance" in regulation is not uniform: it may refer to a single molecule, a mixture of unknown or variable composition (UVCB), a group of related compounds (e.g. PFAS), or even an analytical parameter defined as a function over multiple substances. As illustrated in Figure 1, these entities correspond to fundamentally different semantic types.

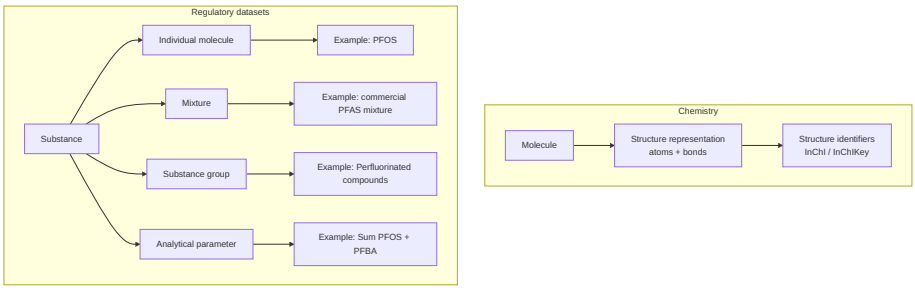

Figure 1. Conceptual model of the semantic mismatch. In chemistry, a substance typically corresponds to a molecule defined by its molecular structure and uniquely identified using structure-derived identifiers such as InChI or InChIKey. In regulatory datasets, however, the concept of a "substance" often encompasses heterogeneous entities including individual molecules, mixtures, substance groups, and analytical sum parameters. These entities correspond to fundamentally different semantic types, which complicates interoperability and automated reasoning across regulatory datasets. This conceptual mismatch is reflected empirically in Section 3.3, where a large fraction of regulatory entries cannot be linked to a defined molecular structure.

This mismatch has profound implications for interoperability. Because regulatory substance definitions lack formal semantics and structure-based identifiers, the relationships between substances across datasets—such as equivalence, subset relations, or partial overlap—are rarely explicit, in particular for group- and parameter-based substances as illustrated in Figure 2. As a result, integrating substance information across regulatory

frameworks becomes a complex and largely manual task, hindering data reuse, policy alignment, and the implementation of FAIR data principles.

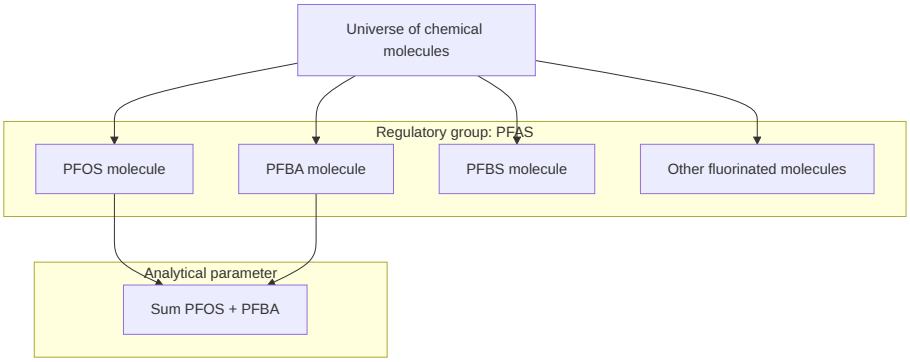

Figure 2. Set-theoretic interpretation of regulatory substances. Many regulatory "substances" correspond not to individual molecules but to sets of molecules or analytical functions defined over such sets. For example, regulatory groups such as PFAS represent collections of structurally related molecules, while analytical parameters such as "Sum PFOS + PFBA" represent measurements derived from subsets of these molecules. Without an explicit formal representation of these set relationships, the semantic relations between regulatory entities remain undefined, complicating interoperability and automated reasoning across datasets. The absence of explicit set-based semantics for such entities explains the combinatorial complexity observed in Section 3.7.

The problem is further exacerbated by the scale and diversity of European regulatory data. Multiple overlapping lists of substances of concern exist, each developed within a specific legal and domain context. Without a shared representation of chemical identity, these lists cannot be reliably linked or compared. Attempts to resolve these inconsistencies post hoc—whether through manual curation or data-driven methods such as text embeddings—face fundamental limitations.

This paper investigates the extent and consequences of this interoperability problem using a comprehensive analysis of 18 regulatory datasets from the European Chemicals Agency (ECHA) and related sources. We focus on three research questions:

- To what extent can regulatory substances be linked to unambiguous chemical structures?
- How consistent and reliable are commonly used identifiers such as CAS numbers for cross-dataset linking?
- Can semantic or data-driven methods compensate for the absence of structure-based identifiers?

Based on this analysis, we make the following contributions:

1. We quantify the heterogeneity of regulatory substance definitions and introduce a taxonomy of linkability reflecting their degree of chemical specificity.
2. We demonstrate the limitations of CAS-based identification and show that a substantial fraction of regulatory substances cannot be resolved to a defined structure.

3. We evaluate semantic embedding approaches and show that they fail to recover chemical identity for non-structure substances.
4. We estimate the scale of the manual effort required for retrospective harmonisation, demonstrating its infeasibility.
5. We show that structure-based identifiers combined with ontology-driven approaches enable scalable integration through knowledge graphs.

Our findings indicate that interoperability of chemical substance data cannot be achieved solely through technical post-processing. Instead, it must be addressed at the level of data modelling and legislation, by adopting structure-based identifiers, formal ontologies, and linked data principles from the outset.

1.1. Related Work

Interoperability of scientific data has been extensively studied in the context of FAIR principles [3] and knowledge graph infrastructures. In the life sciences, large-scale resources such as PubChem [4] and the Chemical Entities of Biological Interest (ChEBI) ontology [5] demonstrate how chemical entities can be represented using structure-based identifiers, ontology-driven classifications, and machine-readable knowledge graphs. The most extensive work on rule-based chemical classification has been undertaken by ClassyFire [6]. In this framework, a structure-based taxonomy (ChemOnt) has been established, where each class is characterised by precise and unambiguous formal rules. These infrastructures enable consistent integration, querying, and reuse of chemical information across domains.

Within the Semantic Web community, numerous approaches have been proposed for data integration based on linked data principles, ontology alignment, and entity resolution [6]. Techniques such as identifier mapping, schema matching, and, more recently, embedding-based similarity methods [7] have been applied to address heterogeneity across datasets. However, these approaches typically assume that entities correspond to well-defined and comparable concepts.

In the regulatory domain, previous work has highlighted challenges related to data fragmentation, heterogeneity of identifiers, and limited machine-readability of legal data [8]. In particular, chemical regulation introduces additional complexity due to the use of substance groups, mixtures, and analytical parameters that do not correspond to single chemical structures. While prior studies have focused on improving data accessibility or applying FAIR principles to regulatory datasets, the fundamental mismatch between chemical identity and regulatory substance definitions has received limited attention.

This paper builds on these strands of work by showing that interoperability challenges in regulatory chemical data cannot be resolved solely through technical integration methods. Instead, they arise from the absence of a shared, structure-based representation of chemical identity, requiring alignment between domain semantics, data models, and regulatory definitions.

The remainder of this paper is structured as follows. Section 2 describes the data sources and analytical methods. Section 3 presents the empirical findings. Section 4 discusses the implications for regulatory data infrastructures and semantic interoperability. Section 5 concludes with recommendations for future work.

2. Methods

2.1. Data Integration

Substance records were compiled from 18 European regulatory sources, including REACH-related lists (candidate, authorisation, restriction, POPs [9], CLP), the EU Pesticides Database, and a Flemish administrative list. For each entry, available identifiers (CAS number, EC number, substance name) were collected.

Where possible, substances were linked to structure-based identifiers (InChIKey) via PubChem [4]. The resulting dataset links each regulatory entry to its identifiers and source list.

2.2. Chemical Identity and Linkability

Each substance entry was classified into one of several types reflecting its degree of chemical specificity: structure-defined molecule, substance group, mixture/UVCB, analytical parameter, or administrative entry.

Based on this classification, a linkability taxonomy was defined, ranging from fully structure-resolvable entries (InChIKey) to non-identifiable entries. This taxonomy expresses the extent to which regulatory substances can be unambiguously linked across datasets.

2.3. Identifier Consistency Analysis

For entries with both CAS numbers and InChIKeys, bidirectional mappings were analysed to assess identifier consistency. This analysis evaluates whether CAS numbers uniquely map to structures and vice versa.

2.4. Structure Coverage and Cross-List Overlap

For each regulatory source, the proportion of structure-defined substances was calculated. Overlap between sources was analysed using structure-based matching (InChIKey) and set similarity measures.

2.5. Analysis of Non-Structure Substances

Substances without structure identifiers were analysed using text-based methods, including embedding-based clustering [7,10,11,12,13] and similarity matching to chemical taxonomies (ChemOnt), to assess whether semantic techniques can recover meaningful structure or classification [14,6] .

2.6. Knowledge Graph Construction

All data were integrated into an RDF knowledge graph using SKOS [15] for taxonomy representation. Structure-defined substances were linked to ChemOnt classes and to ChEBI entities via InChIKey, enabling integration of biological hazard annotations [16].

2.7. Workload Estimation

For non-structure substances, interoperability requires manual assessment of pairwise relationships (equivalence, subset, overlap, disjoint). The required effort scales quadratically with the number of such entries as $\binom{N}{2} = N(N-1)/2$ where N is the number of non-structure entries under consideration.

2.8. Priority Scoring

To illustrate the analytical potential enabled by the integrated knowledge graph, a composite priority score was defined by combining two independent dimensions: (i) regulatory presence across multiple European substance lists (e.g. CLP [2], EU Chemicals Strategy for Sustainability [17], and POPs [9]), and (ii) biological role annotations derived from ChEBI [5].

The resulting score does not represent a formal risk assessment, but serves as an illustrative aggregation of heterogeneous regulatory and biological information within a unified semantic framework. Its purpose is to demonstrate how interoperable, structure-linked data enables cross-domain queries and comparative analyses that are not feasible in fragmented regulatory datasets.

This approach highlights how combining regulatory metadata with ontology-based annotations can support the identification of substances and chemical classes that are prominent across multiple dimensions of interest.

3. Results

3.1. Overview of the integrated knowledge graph

We constructed an integrated knowledge graph to analyse the representation and interoperability of regulatory substances across datasets. Figure 3 provides an overview of the resulting graph structure and its main semantic layers.

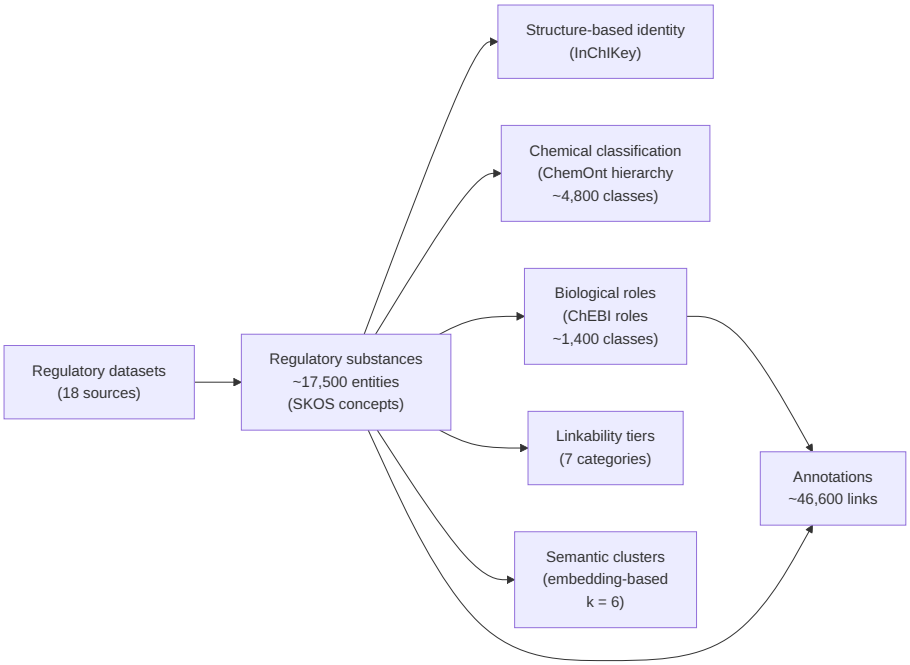

Figure 3. Overview of the integrated knowledge graph constructed from 18 regulatory datasets. Regulatory substances (17,500 entities) are represented as SKOS concepts and form the core layer of the graph. Where possible, substances are linked to structure-based identifiers (InChIKey), enabling alignment with chemical classification systems (ChemOnt) and biological roles (ChEBI). In addition, substances are organised into linkability tiers reflecting their degree of chemical specificity, and into semantic clusters derived from embedding-based similarity. The graph is further enriched with annotations (46,600 links) connecting substances to regulatory sources and functional roles. The figure highlights the multiple semantic layers required to represent and integrate heterogeneous regulatory substance definitions.

3.2. Dataset Overview

Integration of 18 regulatory sources yielded 41,813 records. Of these, 63% could be linked to a defined molecular structure via InChIKey.

3.3. Heterogeneity of Substance Definitions

A substantial fraction of regulatory entries does not correspond to a well-defined chemical structure. Approximately 37% of entries are non-structure substances, including mix-

tures (UVCBs), substance groups, analytical parameters, or entries lacking usable identifiers. This heterogeneity reflects fundamental differences in how "substances" are defined across regulatory contexts (cf. Figure 2).

The degree of heterogeneity varies strongly between regulatory instruments: some lists are largely structure-based, while others are dominated by group or mixture definitions, limiting their interoperability.

Application of the seven-tier linkability taxonomy (Section **??**) to the full dataset reveals that 20,909 entries (62.5%) are fully addressable via a direct InChIKey (tier 1; Figure 4). A further 60 entries correspond to substance groups that can be partially resolved: 15 entries (<0.1%) belong to tier 2 (group with resolvable base compound) and 45 (0.1%) to tier 3 (group with unresolvable base compound). Together, these tiers represent the structure-addressable portion of the dataset.

The remaining 37.5% of entries are not directly addressable by structure. Among these, 3,865 entries (11.6%) carry a CAS number but cannot be resolved to a unique InChIKey (tier 4), demonstrating that CAS identifiers do not guarantee structural identity. A further 3,622 entries (10.8%) correspond to UVCB substances, undefined mixtures, or analytical parameters (tier 5), for which no single molecular representation exists. Additionally, 15 entries (<0.1%) represent administrative or regulatory constructs (tier 6), and 4,981 entries (14.9%) lack any actionable identifier (tier 7). This final group — nearly one in six regulatory records — constitutes the most significant barrier to automated cross-dataset substance matching.

These results show that more than one-third of regulatory substance entries cannot be linked to a well-defined chemical structure, and that even commonly used identifiers such as CAS numbers fail to ensure unambiguous identification. As a result, structure-based integration approaches can only be applied to a subset of regulatory data, while the remainder requires fundamentally different modelling strategies.

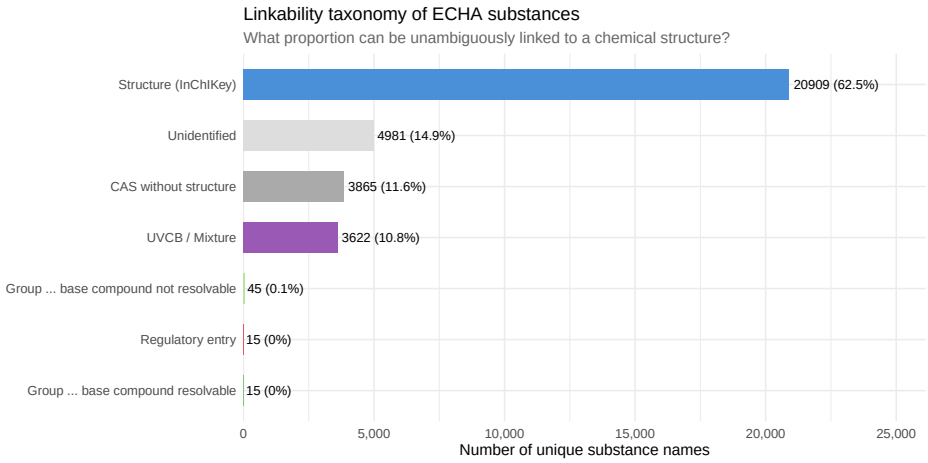

Figure 4. Seven-tier linkability taxonomy applied to all 33,452 unique regulatory substance entries. Tier 1 (62.5%) corresponds to substances that can be directly linked to a defined chemical structure via InChIKey. The remaining tiers represent decreasing levels of structural specificity, including group definitions, mixtures, and entries lacking actionable identifiers. Overall, 37.5% of regulatory substances cannot be resolved to a unique chemical structure.

3.4. Limitations of CAS Numbers

Although CAS numbers are widely used, they do not provide a reliable unique identifier. Multiple CAS numbers may refer to the same structure, and conversely, a single CAS number may correspond to multiple structures. This confirms that CAS-based linking is insufficient for interoperability.

3.5. Limited Interoperability Across Regulatory Lists

Cross-list analysis shows that most substances are unique to a single regulatory source. Overlap between lists is limited, reflecting the domain-specific design of regulatory instruments rather than a shared chemical reference framework.

3.6. Failure of Semantic Approaches for Non-Structure Substances

Embedding-based clustering of non-structure substance names yields semantically coherent groups, but does not resolve chemical identity. While clusters reflect lexical or contextual similarity between substance descriptions, they do not correspond to well-defined molecular entities.

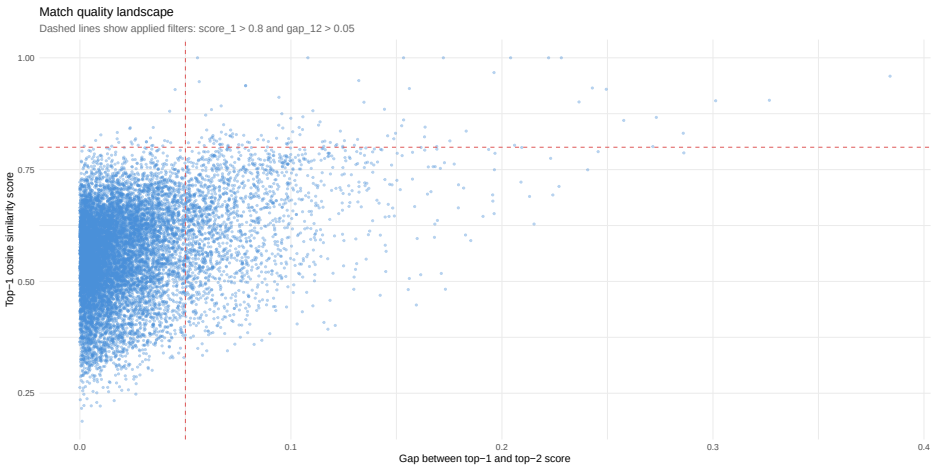

Figure 5. Match quality landscape: score gap Δ_{12} versus best match score s_1 for all non-structure substance–ChemOnt class pairs. Entries in the upper-right region satisfy both quality thresholds and are retained as high-confidence matches.

To assess whether semantic similarity can be used to recover chemical meaning, we attempted to match non-structure substances to chemical taxonomy classes (ChemOnt). High-confidence matches were obtained for less than 1% of cases, indicating that semantic approaches are largely insufficient for reconstructing chemical identity.

The match quality landscape (Figure 5) plots the score gap Δ_{12} between the best and second-best matches against the best match score s_1 for all substance–class pairs. Entries in the upper-right region (high s_1, high Δ_{12}) correspond to unambiguous matches and

represent the 107 high-confidence mappings retained in our analysis. In contrast, entries with high s_1 but low Δ_{12} indicate genuine ambiguity between multiple candidate classes, reflecting the underspecified nature of the original substance definitions.

The predominance of low-confidence and ambiguous matches demonstrates that semantic and embedding-based methods cannot compensate for the absence of structure-based identifiers. In particular, these approaches fail systematically for substance groups, mixtures, and analytical parameters, where no single underlying chemical structure exists.

3.7. Scale of the Interoperability Problem

For the more than 15,000 non-structure entries, full manual reconciliation would require on the order of hundreds of millions of pairwise comparisons. Under conservative assumptions, this corresponds to several thousand person-years of expert effort.

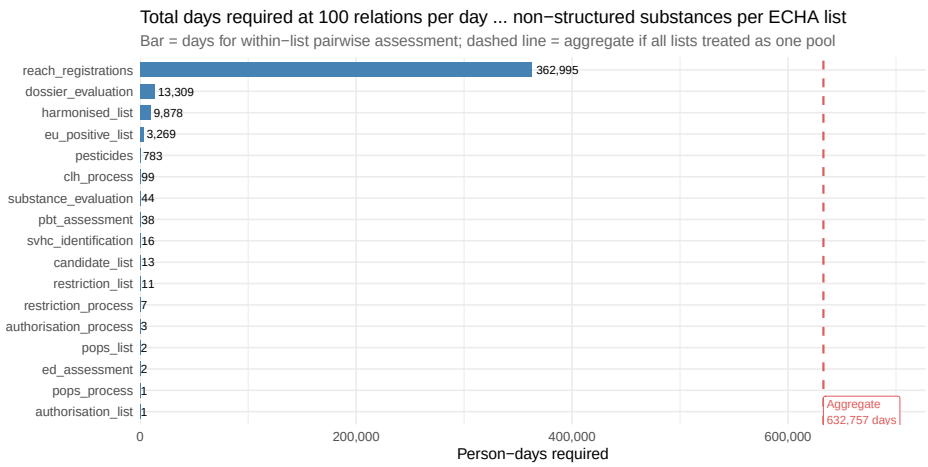

Figure 6. Estimated pairwise curation workload for 15,432 non-structure entries across the combined regulatory dataset, broken down by source. The total corresponds to approximately 1.2 million person-days (nearly 6,000 person-years) under an assumed throughput of 100 pair assessments per person-day. The quadratic growth of the workload illustrates that manual reconciliation is not scalable, reinforcing the need for structure-based and ontology-driven representations at the source.

Figure 6 quantifies this workload across all regulatory sources, illustrating both the absolute scale and the quadratic growth that makes each additional non-structure entry disproportionately costly. Even moderate increases in the number of unresolved entries lead to a rapidly escalating curation burden.

Manual pairwise reconciliation at this scale is not operationally feasible. This result highlights a fundamental constraint: interoperability cannot be achieved through retrospective harmonisation of heterogeneous substance definitions.

Instead, the only scalable strategy is to address the problem at its source. By requiring structure-resolvable identifiers and ontology-based representations in regulatory definitions wherever chemically applicable, the number of entries requiring manual reconciliation can be structurally reduced rather than managed post hoc.

3.8. *Knowledge Graph Integration*

Structure-defined substances can be successfully integrated across regulatory lists, chemical taxonomies (ChemOnt), and biological annotations (ChEBI) using InChIKey as a linking key. This demonstrates the feasibility of a linked-data approach when structure information is available.

3.9. *Priority Patterns*

The integrated knowledge graph enables the combination of regulatory and biological annotations into a shared analytical space. Using the composite priority score defined in Section 2.8, we identify substances and chemical classes that are prominent across both regulatory presence and biological role annotations.

The bubble chart (Figure 7) visualises this priority space by plotting list_score against bio_score for all scored substances. Bubble size is proportional to the number of regulatory source memberships.

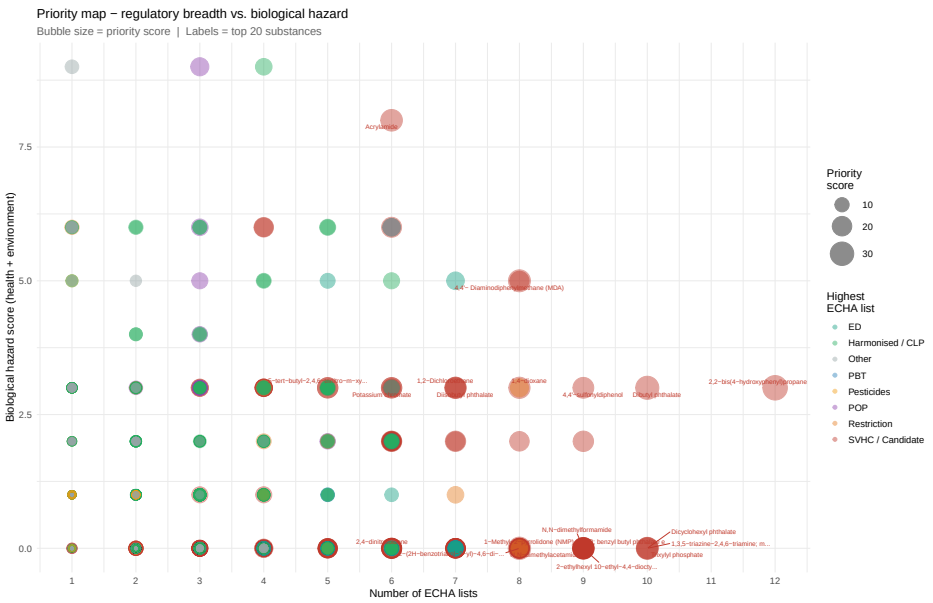

Figure 7. Bubble chart of composite priority space: regulatory list score versus biological hazard score for all scored substances. Bubble size is proportional to the number of regulatory source memberships. Substances in the upper-right quadrant combine broad regulatory coverage with documented biological hazard evidence.

Substances located in the upper-right quadrant combine high regulatory coverage with multiple biological role annotations, reflecting their prominence across heterogeneous data sources. Examples include bisphenol A (BPA) and dibutyl phthalate, which appear consistently across regulatory lists and are associated with multiple biological roles in ChEBI.

Importantly, this analysis does not constitute a toxicological ranking, but demonstrates how interoperable data representations enable the integration of regulatory and ontology-based information within a single analytical framework.

At the class level, a small number of chemical classes accounts for a large fraction of regulated substances. This concentration suggests that class-based representations can provide a scalable bridge between individual substances and regulatory group definitions, supporting alignment across regulatory contexts.

4. Discussion

4.1. Chemical identity as a prerequisite for interoperability

The results demonstrate that interoperability of regulatory substance data is fundamentally constrained by the absence of a shared and formalised notion of chemical identity. While structure-based identifiers such as InChIKey enable unambiguous linking across datasets, a substantial fraction of regulatory entries cannot be expressed in this form. Instead, regulatory "substances" frequently represent heterogeneous entities, including mixtures, substance groups, and analytical parameters.

This heterogeneity is not merely a data quality issue but reflects a structural mismatch between chemical reality and regulatory abstraction. As a consequence, semantic relationships between regulatory entries—such as equivalence, subset relations, or partial overlap—remain implicit and undefined. Without explicit formalisation of these relationships, automated integration across regulatory datasets is inherently limited.

4.2. Limitations of identifier-based approaches

The analysis confirms that commonly used identifiers in regulatory contexts, particularly CAS numbers, do not provide a reliable foundation for interoperability. Although widely adopted, CAS numbers exhibit both one-to-many and many-to-one mappings with chemical structures, reflecting differences in stereochemistry, salt forms, and context-dependent definitions.

This finding challenges a widespread implicit assumption in regulatory data integration: that CAS numbers can serve as globally unique identifiers. In practice, reliance on CAS numbers introduces ambiguity rather than resolving it, undermining their suitability as a backbone for cross-dataset linking.

4.3. Why semantic and AI-based approaches are insufficient

A key question is whether modern data-driven techniques can compensate for the absence of structure-based identifiers. The results indicate that this is not the case. While embedding-based methods can cluster non-structure substance names into semantically coherent groups, they fail to recover chemical identity or establish precise relationships between entities. Mapping to chemical taxonomies such as ChemOnt is only successful in a negligible fraction of cases.

This limitation is structural rather than technical. Textual descriptions of mixtures, substance groups, and analytical parameters do not encode sufficient information to reconstruct the underlying chemical entities. As a result, no post hoc semantic processing

can replace the explicit representation of chemical identity at the source. Embeddings cannot compensate for missing structure.

4.4. The combinatorial barrier to retrospective harmonisation

The scale of the interoperability problem becomes evident when considering the effort required for manual reconciliation. For non-structure substances, establishing semantic relationships requires pairwise comparison, leading to quadratic growth in workload. The estimated effort of several thousand person-years demonstrates that retrospective harmonisation is not a viable strategy at scale.

This reinforces a key insight: interoperability cannot be achieved by correcting data after publication, but must be designed into regulatory data models from the outset.

4.5. Towards ontology-driven regulatory data infrastructures

In contrast to regulatory datasets, cheminformatics infrastructures such as PubChem and ChEBI demonstrate that large-scale interoperability is achievable when chemical identity is explicitly represented and linked to formal ontologies. Structure-based identifiers, combined with hierarchical classifications and machine-readable knowledge graphs, enable consistent integration across domains.

The priority scoring example illustrates that once chemical identity is formalised and linked across datasets, higher-level aggregations combining regulatory and biological information become straightforward to compute.

The results of this study show that similar principles can be applied to regulatory data. Structure-defined substances can be successfully linked across datasets and enriched with chemical and biological annotations when represented using InChIKey and ontology-based classifications. Moreover, the observed concentration of substances within a limited number of chemical classes suggests that class-level representations can provide a scalable bridge between individual molecules and regulatory group definitions.

4.6. Implications for FAIR chemical data in Europe

The findings have direct implications for the implementation of FAIR (Findable, Accessible, Interoperable, Reusable) principles in European chemical regulation. Current regulatory datasets fall short of interoperability primarily because they lack globally unique, persistent, and machine-readable identifiers grounded in chemical structure.

Achieving FAIR chemical substance data requires a shift from document-centric and identifier-fragmented approaches towards linked data infrastructures in which substances are represented as resolvable URIs, grounded in structure-based identifiers and connected through formal ontologies. Importantly, this shift is not purely technical: it requires alignment between data modelling practices and the legal definitions of substances in regulatory frameworks.

4.7. Limitations and scope

This study focuses on datasets derived from ECHA and related European regulatory sources, and on the use of InChIKey as the primary structure-based identifier. While the

results are consistent across the analysed datasets, further work is needed to assess the generality of these findings across other regulatory domains and jurisdictions.

Furthermore, not all regulatory concepts can or should be reduced to individual molecular structures. Mixtures, UVCBs, and analytical parameters are legitimate and necessary constructs in regulation. However, their current representation lacks formal semantics, limiting their integration with structure-based systems. Future work should explore hybrid models that explicitly represent these entities as sets, classes, or functions over well-defined chemical structures.

4.8. From data integration to regulatory design

Taken together, the results show that the interoperability problem is not primarily a technical integration challenge, but a consequence of how substances are defined and represented in regulation. Efforts to improve interoperability must therefore extend beyond data engineering to include the design of regulatory concepts and data standards.

Embedding structure-based identifiers, ontology references, and machine-readable semantics directly into regulatory processes would significantly reduce ambiguity and enable scalable integration. Without such changes, the gap between regulatory data and scientific data infrastructures will continue to widen, limiting the effectiveness of data-driven environmental governance.

Concretely, this implies that regulatory data infrastructures should: (i) include structure-based identifiers (e.g. InChIKey) wherever applicable; (ii) represent substance groups and mixtures using explicit semantic models (e.g. set- or class-based representations); (iii) publish substance data as linked data with persistent URIs.

5. Conclusion

This paper has shown that interoperability of chemical substance data in European regulation is fundamentally limited by the absence of a shared and formalised notion of chemical identity. Across 18 regulatory datasets, a substantial fraction of substances cannot be linked to a defined molecular structure, and commonly used identifiers such as CAS numbers do not provide a reliable basis for cross-dataset integration.

We demonstrate that this heterogeneity leads to unresolved semantic relations between regulatory entities and creates a combinatorial barrier to retrospective harmonisation. Data-driven approaches, including semantic embeddings, are insufficient to recover chemical identity in the absence of structure-based representations.

At the same time, our results show that interoperability becomes feasible when structure-based identifiers such as InChIKey are combined with ontology-driven approaches and knowledge graph representations. This highlights that the interoperability problem is not primarily technical, but originates in how chemical substances are defined and represented in regulatory frameworks.

We therefore argue that achieving FAIR and interoperable chemical data in Europe requires a shift towards structure-based identifiers, formal ontologies, and linked data principles embedded directly in regulatory data models and legislation. Without such changes, the gap between regulatory data and scientific data infrastructures will persist, limiting the effectiveness of data-driven environmental policy.

Future work should focus on developing hybrid semantic models for mixtures and substance groups, as well as on translating these principles into operational standards for regulatory data publication.

Data Availability

All data, code, and materials supporting this study are publicly available. The complete dataset, including the integrated knowledge graph, analysis scripts, and figures, is archived on Zenodo at `https://doi.org/10.5281/zenodo.xxxxxxx`.

The development version of the repository is available on GitHub: `https://github.com/gezever/A-Substance-Is-Not-Always-a-Substance/`.

Machine-readable metadata describing the dataset is provided using the DCAT standard. All generated data products are published as machine-readable RDF and described using DCAT metadata to support FAIR principles.

Supplementary Materials

Supplementary materials include:

- The integrated knowledge graph (RDF/Turtle)
- Processed datasets derived from 18 regulatory sources
- Analysis scripts and workflows
- All figures and visualisations
- DCAT metadata describing the dataset and its distributions
- The initial methods and results section that was to large for this paper

All materials are available via the Zenodo archive associated with this publication.

References

[1] European Parliament, Council of the European Union. Regulation (EC) No 1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH). Official Journal of the European Union; 2006. L 396. Available from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02006R1907-20220301.

[2] European Parliament, Council of the European Union. Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures (CLP). Official Journal of the European Union; 2008. L 353. Available from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02008R1272-20231101.

[3] Wilkinson MD, Dumontier M, Aalbersberg IJ, Appleton G, Axton M, Baak A, et al. The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data. 2016 Mar;3(1):160018. Available from: https://doi.org/10.1038/sdata.2016.18.

[4] Kim S, Thiessen PA, Bolton EE, Bryant SH. PUG-SOAP and PUG-REST: web services for programmatic access to chemical information in PubChem. Nucleic acids research. 2015 Jul;43(W1):W605-11. Available from: https://doi.org/10.1093/nar/gkv396.

[5] Degtyarenko K, de Matos P, Ennis M, Hastings J, Zbinden M, McNaught A, et al. ChEBI: a database and ontology for chemical entities of biological interest. Nucleic Acids Research. 2008 Jan;36(suppl_1):D344-50. Available from: https://doi.org/10.1093/nar/gkm791.

[6] Djoumbou Feunang Y, Eisner R, Knox C, Chepelev L, Hastings J, Owen G, et al. ClassyFire: automated chemical classification with a comprehensive, computable taxonomy. Journal of Cheminformatics. 2016 Nov;8(1):61. Available from: https://doi.org/10.1186/s13321-016-0174-y.

[7] Reimers N, Gurevych I. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics; 2019. p. 3982-92.

[8] Filtz E, Kirrane S, Polleres A. The linked legal data landscape: linking legal data across different countries. Artificial Intelligence and Law. 2021 Dec;29(4):485-539. Available from: https://doi.org/10.1007/s10506-021-09282-8.

[9] European Parliament, Council of the European Union. Regulation (EU) 2019/1021 on persistent organic pollutants (POPs). Official Journal of the European Union; 2019. L 169. Available from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32019R1021.

[10] Wang W, Wei F, Dong L, Bao H, Yang N, Zhou M. MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. In: Advances in Neural Information Processing Systems. vol. 33; 2020. p. 5776-88. Available from: https://proceedings.neurips.cc/paper/2020/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

[11] Rousseeuw PJ. Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. Journal of Computational and Applied Mathematics. 1987;20:53-65.

[12] Lloyd SP. Least Squares Quantization in PCM. IEEE Transactions on Information Theory. 1982;28(2):129-37.

[13] McInnes L, Healy J, Melville J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. arXiv preprint arXiv:180203426. 2018. Available from: https://arxiv.org/abs/1802.03426.

[14] Manning CD, Raghavan P, Schütze H. Introduction to Information Retrieval. Cambridge: Cambridge University Press; 2008.

[15] Miles A, Bechhofer S. SKOS Simple Knowledge Organization System Reference. World Wide Web Consortium; 2009. Available from: https://www.w3.org/TR/skos-reference/.

[16] Sanderson R, Ciccarese P, Young B. Web Annotation Data Model. World Wide Web Consortium; 2017. Available from: https://www.w3.org/TR/annotation-model/.

[17] European Commission. Chemicals Strategy for Sustainability Towards a Toxic-Free Environment. European Commission; 2020. COM(2020) 667 final. Available from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020DC0667.